



KTRADE SECURITIES LIMITED

TO WHOM IT MAY CONCERN

This letter is to certify that Mr. Noman Ali was employed with **KTrade Securities Limited** as a Software Engineer from August 05, 2021 to 4 October 2023, completing approximately 2 year and 2 months of professional service with the organization.

As a **Software Engineer**, Noman demonstrated strong expertise in software engineering, microservices architecture, cloud computing, real-time distributed systems, FinTech, and artificial intelligence.

A significant part of his work involved the development and support of a **virtual trading platform**, including the integration of real-time market data from the **Pakistan Stock Exchange (PSX)** and **Saudi Exchange (Tadawul)**. He contributed to event-driven data pipelines using Redis Streams, processing and distributing real-time market information across distributed microservices and application layers.

His key responsibilities included:

- Designing and developing .NET-based microservices and RESTful APIs
- Implementing distributed and event-driven application components
- Integrating Redis Streams and PostgreSQL
- Working with AWS, Amazon EKS, and Kubernetes-based deployments
- Supporting production systems, deployments, troubleshooting, and optimization
- Collaborating with frontend and backend engineering teams
- Participating in technical architecture, code reviews, and mentoring

In addition, Noman contributed to a **Proof of Concept involving Quantitative Development and Artificial Intelligence**, with a focus on Reinforcement Learning and its potential applications in financial and trading environments.

MAIN OFFICE

101-102, 1st Floor, Plot# 33-C,
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Based on his performance and contributions, we consider Noman a **technically capable and versatile software engineering professional** with valuable experience in complex production systems and FinTech technologies.

We wish him continued success in his future professional endeavours and are pleased to provide this reference for his professional and career-related purposes.

Sincerely,
For KTrade Securities Limited

Mahmood Ali Shah Bukhari
Chief Executive Officer

Issue Date: **September 03, 2026**

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